JOEL MILLA

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EDUCATION

Carnegie Mellon University **Jun 2028**
Master of Science in Robotic Systems Development
- **Coursework**: Robot Autonomy; Computer Vision; Robot Mobility on Air, Land, Sea; Systems Engineering.

Instituto Tecnológico y de Estudios Superiores de Monterrey **Jul 2026**
Bachelor of Computing in Computer Science and Technology
- **Academic Achievements**: Top 1% in the School of Engineering, GPA: 98.06/100.

SKILLS

Software: Proficient (4-5 years): C++, Git, Linux, Python || Experienced (2-3 years): ROS2, OpenCV, PCL, Docker, PyTorch
Robotics: Camera calibration, ArUco pose estimation, LiDAR-camera fusion, particle filter localization, NVIDIA Jetson

RESEARCH

Artificial Intelligence and Robotics Laboratory (Tec de Monterrey), *Research Intern* **Jul 2025 - Jan 2026**
- Trained a LaneATT row detection model in PyTorch and deployed it on a Jetson Xavier NX for autonomous greenhouse navigation, building a 3,872-image dataset and reaching 91% F1 score at 48 ms per frame on embedded hardware.
- Built the robot's gimbal-stabilized camera rig and calibrated its intrinsic matrix for ArUco pose estimation, using the poses as ground truth to measure 0.133 m RMS lateral deviation (12.8% of row width) in the field.
- Developed the C++ navigation control system and a ROS2 evaluation pipeline for the robot, validating real-time row following through 50+ greenhouse field deployments.
- Published as co-author: "Efficient Real-Time Row Detection and Navigation Using LaneATT for Greenhouse Environments," Agriculture 16(1), 2026.

WORK EXPERIENCE

Google (New York City, New York), *Software Engineer Intern* **Jun 2023 - Sep 2023**
- Launched a YouTube Search feature to over 2 million users, owning design proposal through C++ and Protocol Buffers implementation, testing, code review, and deployment in Google's large-scale production system.
- Collaborated with a fellow intern to improve search for 14M+ learners struggling with content discovery, reducing query retries among users unable to find relevant educational videos.

PROJECTS

Autonomous Racing F1TENTH Car (Carnegie Mellon University), *Course Project* **Aug 2026 – Oct 2026**
- Developed the autonomy stack for a 1/10 scale F1TENTH car in ROS2, implementing TTC emergency braking, PID wall following, Pure Pursuit with particle filter localization, and an RRT local planner in C++ on an inflated LiDAR occupancy grid.
- Deployed a Follow the Gap planner in Python for reactive obstacle avoidance, tuning speed profiles and steering gains to place 2nd of 12 teams with the fastest qualifying lap (7.55 s) in the CMU Robot Mobility race.

Sensor Fusion Collision Detection System, *Independent Project* **Jun 2025 – Aug 2025**
- Built a C++ multimodal pipeline aligning KITTI camera images and LiDAR point clouds, applying the dataset's extrinsic calibration, rectification, and projection matrices to map 3D LiDAR points into 2D image coordinates.
- Used aligned sensor data to associate LiDAR points with YOLOv3 vehicle detections through bounding-box ROI matching, then estimated the lead vehicle's time-to-collision on each frame of a 10 Hz KITTI sequence.

LiDAR-Based Perception System, *Independent Project* **Jan 2025 - May 2025**
- Built a C++ LiDAR obstacle detection pipeline with RANSAC segmentation and KD-tree Euclidean clustering implemented from scratch, detecting 10 to 17 obstacles per frame across a 22-frame urban driving sequence.
- Reduced processing time from 1.2 s to 0.2 s per scan (6x faster) by downsampling point clouds 85% with voxel filtering and region-of-interest cropping and tracked a cyclist across 24 consecutive frames with PCL's particle filter.

COMPETITIONS

International Collegiate Programming Contest (ICPC), *Contestant* **Aug 2022 - Oct 2024**
- Obtained 64th/482 at Mexico Grand Prix (2024), placed 3rd (2023) and 6th (2022) in University Championship.

HackMTY, *Team Member* **Sep 2024 - Sep 2024**
- Devised a WhatsApp banking system enabling account creation through computer vision authentication and AI-powered recommendations, finishing top 40 among 720+ teams in Latin America's largest student hackathon.